

Cisco Home Lab Configuration Guide

WAN-to-LAN Routing with NAT, DHCP, and ACL

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Home Lab — Networking Practice

Lab Date:

_____ (fill in)

Platform:

Cisco IOS 15.x | Cisco 1941 Router | Catalyst 2960X

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Section 1 — Lab Overview

1. Lab Overview

This lab guides a networking student through the complete configuration of a functional home network using enterprise-grade Cisco equipment. The goal is to establish full internet connectivity from LAN-attached hosts by building a three-device topology: an **Arris TG1682G** cable modem serving as the WAN uplink, a **Cisco 1941 Router** performing Network Address Translation (NAT), acting as a DHCP server for LAN clients, and routing between the WAN and LAN segments, and a **Cisco Catalyst 2960X** switch providing Layer 2 LAN switching for end hosts.

Upon successful completion of this lab, LAN hosts will dynamically receive IP addresses from the router's DHCP pool, route their traffic through the Cisco 1941, have their private IP addresses translated to the router's public WAN address via NAT overload (PAT), and reach the internet. This lab reinforces foundational networking skills including interface configuration, DHCP, NAT/PAT, standard ACLs, and static routing.

Topology Summary

Device	Role	Interface Used
Arris TG1682G	Cable Modem / WAN Gateway	Ethernet LAN port → Gi0/0 of router
Cisco 1941 Router	Gateway Router, NAT, DHCP Server	Gi0/0 (WAN), Gi0/1 (LAN)

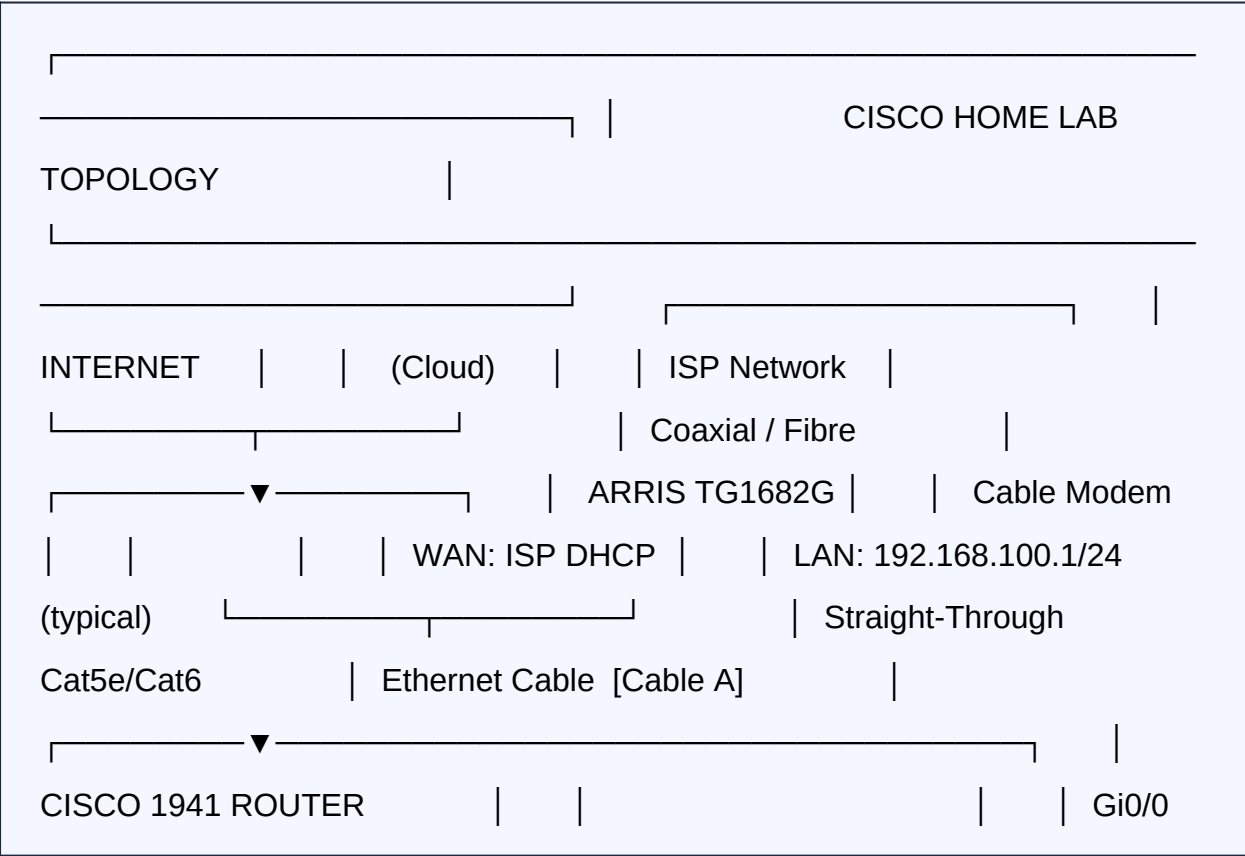
Device	Role	Interface Used
Cisco Catalyst 2960X	Layer 2 LAN Switch	Any access port (to Gi0/1 of router)
PC / Host	LAN Client	Connected to switch access port

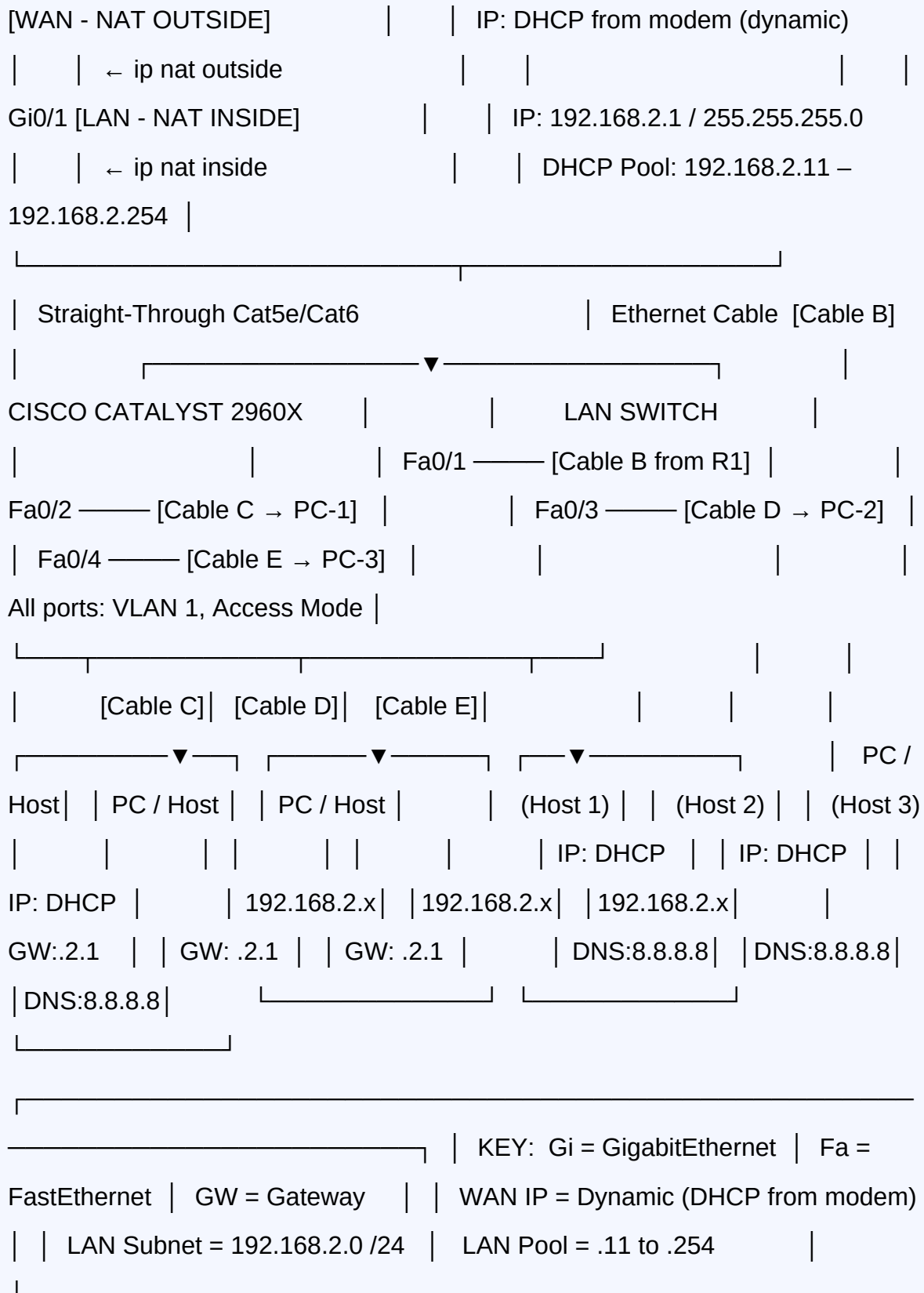
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Section 2 — Topology Diagram

2. Physical Topology Diagram

The diagram below shows the complete physical and logical layout of the lab. Interface labels and IP addresses are shown on each link for reference.





3. Equipment & Prerequisites

3.1 Hardware Required

#	Item	Specification / Notes
1	Arris TG1682G Cable Modem	Or any cable modem with a DHCP-enabled LAN Ethernet port for the WAN uplink
2	Cisco 1941 Router	Must be running Cisco IOS 15.x (Security or Data license). Two built-in GigabitEthernet interfaces required
3	Cisco Catalyst 2960X Switch	Any 24- or 48-port model; no Layer 3 features required for this lab
4	Ethernet Cables (×3 minimum)	Straight-through Cat5e or Cat6; one per link: Modem→Router, Router→Switch, Switch→PC
5	Console Cable	RJ45-to-DB9 rollover cable, or USB-to-RJ45 console cable if using a modern laptop

#	Item	Specification / Notes
6	PC or Laptop	Used both as console terminal (USB/COM port) and LAN client. Windows, macOS, or Linux

3.2 Software Prerequisites

- **Terminal Emulator:** PuTTY (free), SecureCRT, or TeraTerm — configured for serial/COM port access
- **Serial Settings:** Baud Rate: 9600 | Data bits: 8 | Parity: None | Stop bits: 1 | Flow control: None (9600-8-N-1)
- **Cisco IOS Image:** IOS 15.x must be loaded on the 1941 router (verify with `show version`)
- **Driver (if needed):** USB-to-serial driver (e.g., Prolific PL2303 or FTDI) for USB console cable on Windows

TIP — Finding Your COM Port

On Windows: open

Device Manager → Ports (COM & LPT)

to find which COM port is assigned to your console cable (e.g., COM3). Enter that COM number in PuTTY under the Serial connection type.

4. Physical Cabling Steps

Follow these steps in order to correctly cable the lab before applying any software configuration. Incorrect cabling is the most common cause of lab failure.

1. **Power OFF all devices** before making any cable connections. This prevents port damage and avoids accidental spanning-tree issues.
2. Connect the **Arris TG1682G modem's LAN/Ethernet port** → **Cisco 1941 GigabitEthernet0/0** (the WAN interface) using a straight-through Cat5e/Cat6 Ethernet cable. *[Cable A]*
3. Connect **Cisco 1941 GigabitEthernet0/1** (the LAN interface) → **any available access port on the Catalyst 2960X switch** using a straight-through Ethernet cable. *[Cable B]*
4. Connect each **PC/host network interface card (NIC)** → **any available access port on the Catalyst 2960X switch** using straight-through Ethernet cables. *[Cables C, D, E...]*
5. Connect the **console cable** from your PC's USB or COM port to the **Cisco 1941 Console (RJ45) port**. Open your terminal emulator and set the serial parameters (9600-8-N-1).
6. **Power ON the Arris TG1682G modem first.** Wait until the modem has fully synchronized with the ISP — the online/internet status LED should be solid (not blinking). This may take 2-5 minutes.
7. **Power ON the Cisco 1941 Router.** The router will begin its IOS boot sequence. Wait for the command prompt to appear in the terminal emulator (approximately 60-90 seconds).
8. **Power ON the Catalyst 2960X Switch.** Allow time for the switch to complete its boot sequence and spanning-tree convergence.
9. **Power ON the PC(s) / host(s).**

IMPORTANT — Verify Link LEDs Before Proceeding

Before applying any CLI configuration, verify that all physical link indicator LEDs are green

(active link) or

amber

(initializing) on both the switch and router. A dark/off LED indicates no physical link — recheck the cable and port connection on that segment before continuing.

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Section 5 — Router Configuration

5. Cisco 1941 Router Full Configuration

This is the core section of the lab. All CLI commands must be entered on the Cisco 1941 router via the console connection. Work through each sub-section in order — do not skip steps. Commands shown with the full prompt (e.g., R1(config)#) for clarity.

5.1 — Enter Privileged & Global Configuration Mode

From the initial user EXEC prompt, escalate to privileged EXEC mode using `enable`, then enter global configuration mode using `configure terminal`:

```
Router> enable Router# configure terminal Enter configuration commands, one
per line. End with CNTL/Z. Router(config)#
```

5.2 — Set Hostname

Setting a meaningful hostname makes it easy to identify the device in the prompt and in show outputs. The hostname **R1** is used throughout this guide:

```
Router(config)# hostname R1 R1(config)#
```

NOTE

After executing the

hostname

command, the CLI prompt immediately changes to reflect the new name. All
subsequent commands in this guide use the

R1

prompt.

5.3 — Configure WAN Interface (Gi0/0) — DHCP from Modem

GigabitEthernet0/0 is the WAN-facing interface connected to the Arris TG1682G modem. Rather than assigning a static IP, this interface uses `ip address dhcp` to automatically obtain an IP address from the modem's built-in DHCP server — exactly as a home router would. The `ip nat outside` command designates this interface as the NAT "public" side.

```
R1(config)# interface GigabitEthernet0/0 R1(config-if)# description WAN -  
Connected to Arris TG1682G Modem R1(config-if)# ip address dhcp R1(config-  
if)# ip nat outside R1(config-if)# no shutdown R1(config-if)# exit
```

EXPECTED RESULT

After

```
no shutdown
```

, you should see a syslog message such as:

```
%DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/0 assigned DHCP  
address 192.168.100.x
```

(the exact IP depends on your modem). If you do not see this within 30 seconds, verify the physical cable to the modem.

5.4 — Configure LAN Interface (Gi0/1) — Static IP for LAN Gateway

GigabitEthernet0/1 faces the Catalyst 2960X switch and serves as the **default gateway** for all LAN hosts. A static IP of `192.168.2.1/24` is assigned — this address is what LAN hosts will use as their gateway. The `ip nat inside` command designates this interface as the NAT "private" side.

```
R1(config)# interface GigabitEthernet0/1 R1(config-if)# description LAN -  
Connected to Catalyst 2960X Switch R1(config-if)# ip address 192.168.2.1  
255.255.255.0 R1(config-if)# ip nat inside R1(config-if)# no shutdown  
R1(config-if)# exit
```

5.5 — Configure DHCP Pool (HOME-LAN)

The Cisco 1941 router will act as the DHCP server for the LAN subnet. The pool is named **HOME-LAN** and covers the entire `192.168.2.0/24` subnet. Addresses `192.168.2.1` through `192.168.2.10` are excluded from dynamic assignment to preserve them for infrastructure (gateway, printers, servers, etc.). Hosts will receive addresses starting from `192.168.2.11` onwards.

```
R1(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10 R1(config)# ip  
dhcp pool HOME-LAN R1(dhcp-config)# network 192.168.2.0 255.255.255.0  
R1(dhcp-config)# default-router 192.168.2.1 R1(dhcp-config)# dns-server  
8.8.8.8 1.1.1.1 R1(dhcp-config)# lease 1 R1(dhcp-config)# exit
```

DHCP Parameter	Value	Explanation
excluded-address	192.168.2.1 -	Reserved for

DHCP Parameter	Value	Explanation
	192.168.2.10	infrastructure; not assigned to hosts dynamically
network	192.168.2.0 /24	The subnet served by this DHCP pool
default-router	192.168.2.1	Pushed to hosts as their default gateway (router's Gi0/1)
dns-server	8.8.8.8 & 1.1.1.1	Google Public DNS (primary), Cloudflare DNS (secondary)
lease	1 day	DHCP lease duration; hosts renew after 12 hours

5.6 — Configure NAT Overload (PAT)

NAT overload — commonly called **PAT (Port Address Translation)** — allows all LAN hosts sharing the single, dynamically-assigned WAN IP on Gi0/0 to simultaneously access the internet. It accomplishes this by tracking unique port numbers for each session. A standard ACL (Access Control List number 1) defines which source IP addresses are eligible for translation.

Step 1 — Create Standard ACL 1 to Permit LAN Traffic

ACL 1 uses a wildcard mask of 0.0.0.255 to match all hosts in the 192.168.2.0/24 subnet. Only traffic sourced from this subnet will be translated by NAT:

```
R1(config)# access-list 1 permit 192.168.2.0 0.0.0.255
```

Step 2 — Apply NAT Overload Using ACL 1 and the WAN Interface

This single command ties everything together: traffic matched by ACL 1 (LAN hosts in 192.168.2.0/24) will have its source address translated to the dynamic IP on GigabitEthernet0/0. The `overload` keyword enables PAT (many-to-one translation):

```
R1(config)# ip nat inside source list 1 interface GigabitEthernet0/0 overload
```

① HOW ACL 1 WORKS IN THIS CONTEXT

The wildcard mask

0.0.0.255

is the inverse of the subnet mask. A

0

bit means "must match," a

1

bit means "don't care." So

192.168.2.0 0.0.0.255

matches any address from

192.168.2.0

to

192.168.2.255

— the entire /24 subnet. Traffic not matching this ACL is

not translated

and will be dropped at the NAT boundary.

5.7 — Configure Default Route (Optional but Recommended)

A default route tells the router where to send all traffic for destinations not listed in its routing table — in this case, all internet-bound traffic. When `ip address dhcp` is used on Gi0/0, Cisco IOS should automatically install a default route provided the modem includes the router option (Option 3) in its DHCP response. If it does not, add the route manually.

Option A — DHCP-injected default route (preferred):

```
R1(config)# ip route 0.0.0.0 0.0.0.0 dhcp
```

Option B — Manual static default route (use if DHCP does not inject it):

```
R1(config)# ip route 0.0.0.0 0.0.0.0 [modem-gateway-ip]
```

TIP — Finding the Modem's Gateway IP

After Gi0/0 receives its DHCP address, run

```
R1# show ip route
```

to check if a default route (marked

S*

) was automatically installed. If not, run

```
R1# show dhcp lease
```

to find the modem's IP address and use it in Option B above. The modem gateway is typically

192.168.100.1

or

10.0.0.1

.

5.8 — Save Configuration

Always save the running configuration to NVRAM (startup-config) so it persists across reboots. The router will prompt for a destination filename — simply press

Enter to accept the default:

```
R1(config)# end R1# copy running-config startup-config Destination filename  
[startup-config]? [press Enter] Building configuration... [OK] R1#
```

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6. Catalyst 2960X Switch Configuration

For this basic LAN switching lab, the Catalyst 2960X requires very minimal configuration. By default, all switch ports operate in **access mode on VLAN 1** — this means any device plugged into any port can immediately communicate at Layer 2 without any port-level configuration. However, the steps below apply essential security hardening and identity configuration that every network device should have.

6.1 — Basic Switch Setup

```
Switch> enable Switch# configure terminal Switch(config)# hostname SW1
SW1(config)# no ip domain-lookup SW1(config)# enable secret cisco123
SW1(config)# line console 0 SW1(config-line)# password cisco123 SW1(config-
line)# login SW1(config-line)# exit SW1(config)# end SW1# copy running-config
startup-config
```

Command	Purpose
hostname SW1	Sets the switch hostname for identification in the CLI prompt
no ip domain-lookup	Prevents the switch from trying to DNS-resolve mistyped commands (avoids long pauses)
enable secret cisco123	Protects privileged EXEC mode with an MD5-hashed password
line console 0 / password / login	Requires a password to access the switch via console cable
copy running-config startup-config	Saves the configuration to NVRAM so it persists across reboots

6.2 — Verify Switch Port Status

After cabling and powering on, confirm that connected ports are up and MAC addresses are being learned:

```
SW1# show interfaces status SW1# show mac address-table
```

WHAT TO LOOK FOR

show interfaces status

should display ports connected to the router and PCs as
connected

with the correct speed (e.g., 1000-full for GigabitEthernet).

show mac address-table

should show MAC entries for each active device on VLAN 1.

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Section 7 — Verification

7. Verification Commands

Run these verification commands in order from top to bottom after completing all configuration steps. Each command confirms a specific layer of the lab is working correctly. Do not skip ahead — work sequentially, resolving any failures before proceeding.

#	Command	What to Verify	Expected Output / Success Indicator
1	show ip interface brief	All interfaces are up/up	Gi0/0 and Gi0/1 both show up in Status column and up in Protocol column
2	show ip dhcp binding	DHCP leases assigned to hosts	Shows IP address, MAC address, and lease expiration for each connected LAN host (192.168.2.11+)
3	show ip dhcp pool	Pool utilization and statistics	Shows HOME-LAN pool, network 192.168.2.0/24, number of leases given vs. available
4	show ip nat translations	Active NAT translations exist	Shows inside local → inside global → outside global translation entries after hosts generate traffic
5	show ip route	Routing table with default route	Shows c 192.168.2.0/24 (connected) and S* 0.0.0.0/0 (default route via modem)
6	show running-config	Full configuration review	All configured sections present: interfaces, DHCP pool, ACL 1, NAT overload command
7	show access-lists	ACL 1 match counts incrementing	Shows permit 192.168.2.0, wildcard bits 0.0.0.255 with a match count greater than 0

#	Command	What to Verify	Expected Output / Success Indicator
8	ping 192.168.2.1 (from host)	LAN gateway reachability	Ping succeeds — replies received, 0% packet loss from any LAN host to the router gateway
9	ping 8.8.8.8 (from router CLI)	WAN / Internet reachability from router	5 out of 5 packets succeed — confirms NAT outside and default route are working
10	ping 8.8.8.8 (from host)	End-to-end NAT translation working	Ping from LAN host reaches internet — confirms full NAT/PAT path is functional

Full Verification Command Block

Run all verification commands in this order on the router CLI:

```
R1# show ip interface brief R1# show ip dhcp binding R1# show ip dhcp pool
R1# show ip nat translations R1# show ip route R1# show running-config R1#
show access-lists R1# ping 192.168.2.1 R1# ping 8.8.8.8
```

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Section 8 — Final Configuration

8. Full Working Lab: Final Configuration Summary

The block below is the complete, clean Cisco IOS running-config for the **Cisco 1941 Router (R1)** as it should appear after successfully completing all steps in Section 5. Use this as your reference to verify your configuration with `show running-config`. Non-relevant default entries have been omitted for clarity.

```
! ! ===== ! Cisco
Home Lab - Final Configuration Summary ! Device : Cisco 1941 Router !
Hostname: R1 ! Revision: 1.0 | Author: Surya ! Purpose : WAN-to-LAN Routing
with NAT, DHCP, and ACL !
===== ! hostname
R1 ! no ip domain lookup ! ! — WAN Interface (NAT Outside)
————— ! interface GigabitEthernet0/0 description
WAN - Connected to Arris TG1682G Modem ip address dhcp ip nat outside no
shutdown ! ! — LAN Interface (NAT Inside) ————— !
interface GigabitEthernet0/1 description LAN - Connected to Catalyst 2960X
Switch ip address 192.168.2.1 255.255.255.0 ip nat inside no shutdown ! !
— DHCP Server Configuration ————— ! ip dhcp
excluded-address 192.168.2.1 192.168.2.10 ! ip dhcp pool HOME-LAN network
192.168.2.0 255.255.255.0 default-router 192.168.2.1 dns-server 8.8.8.8
1.1.1.1 lease 1 ! ! — Standard ACL — Defines NAT-eligible LAN traffic
————— ! access-list 1 permit 192.168.2.0 0.0.0.255 ! ! — NAT Overload
(PAT) ————— ! ip nat inside source list 1
interface GigabitEthernet0/0 overload ! ! — Default Route
————— ! ip route 0.0.0.0 0.0.0.0
dhcp ! end
```

QUICK VERIFICATION

After pasting or confirming the config, run

```
R1# show running-config
```

and visually match each block above. Then run the verification sequence from Section 7. If all 10 checks pass — your lab is complete!

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Section 9 — Troubleshooting

9. Troubleshooting Guide

If any verification check from Section 7 fails, use this table to diagnose and resolve the most common issues. Work from the physical layer upward — always verify cabling and interface status before diving into software configuration.

Symptom	Likely Cause	Recommended Fix
Gi0/0 shows down/down	Ethernet cable not connected, wrong port, or modem is offline	Check the physical cable between the modem's LAN port and router Gi0/0. Verify the modem is powered on and the cable/internet LED is solid
Gi0/0 shows up/up but no DHCP IP assigned	Modem DHCP not responding, or interface not re-requesting DHCP	Enter interface Gi0/0, run <code>no ip address</code> then <code>ip address dhcp</code> to force a DHCP renew. Alternatively, run <code>R1# clear ip dhcp binding *</code> and check modem DHCP server settings
LAN hosts not receiving an IP address	DHCP pool misconfigured, Gi0/1 is down, or excluded range covers all addresses	Verify <code>ip dhcp pool HOME-LAN</code> is correctly configured; confirm Gi0/1 is no shutdown and shows up/up; ensure excluded-address range does not cover the entire /24
Hosts can ping 192.168.2.1 but cannot reach internet	NAT not applied on correct interfaces, or no default route in routing table	Run <code>show ip interface brief</code> — verify Gi0/0 has <code>ip nat outside</code> and Gi0/1 has <code>ip nat inside</code> . Run <code>show ip route</code> and confirm <code>S* 0.0.0.0/0</code> is present
<code>show ip nat</code> translations is empty after host traffic	ACL 1 not matching traffic, NAT overload command missing, or inside/outside not set	Re-verify <code>ip nat inside source list 1 interface GigabitEthernet0/0 overload</code> exists in running-config. Run <code>show access-lists</code> to check

Symptom	Likely Cause	Recommended Fix
		ACL 1 match count. Ensure both interfaces have nat inside/outside configured
IP pings work but DNS resolution fails on hosts	Incorrect or missing DNS servers pushed by DHCP	Run show ip dhcp pool and verify dns-server 8.8.8.8 1.1.1.1 is present. If missing, re-enter the DHCP pool and add the dns-server line. Renew DHCP on the host (ipconfig /renew on Windows)
Console no output in terminal emulator	Wrong COM port, wrong baud rate, or bad console cable	Confirm the correct COM port in Device Manager. Verify PuTTY settings: Speed = 9600, Data bits = 8, Stop bits = 1, Parity = None, Flow control = None. Try pressing Enter a few times after connecting

GENERAL TROUBLESHOOTING TIP — OSI Bottom-Up Approach

Always troubleshoot from Layer 1 (physical) upward. Confirm LEDs are active → interface is up/up → correct IP addressing → routing table correct → NAT translations forming. Skipping layers wastes time. Use

debug ip dhcp server events

and

debug ip nat

sparingly — remember to

undebug all

when done.

10. Lab Completion Checklist

Use this checklist to confirm every major milestone before declaring the lab complete. Check each item off as you verify it. All 12 items must pass for a fully functional lab.

- Modem (Arris TG1682G) is online and fully synchronized with the ISP
- Gi0/0 is **up/up** and a DHCP IP address has been assigned by the modem
- Gi0/1 is **up/up** with static IP `192.168.2.1 /24` configured
- DHCP pool **HOME-LAN** is configured with the correct network, gateway, DNS, and lease settings
- At least one LAN host has successfully received a dynamic IP address from the `192.168.2.11+` pool
- ACL 1 is configured to permit `192.168.2.0` with wildcard `0.0.0.255`
- NAT overload (`ip nat inside source list 1 interface Gi0/0 overload`) is applied
- `show ip nat translations` shows active translation entries after host traffic is generated
- Router can successfully ping `8.8.8.8` from its own CLI (5/5 packets)
- LAN host can successfully ping `8.8.8.8` (internet reachability from a client device)
- LAN host can browse the internet in a web browser (DNS resolution + NAT fully working)
- Configuration saved to startup-config using `copy running-config startup-config`

11. Key Concepts Reference

These reference boxes explain the core networking concepts applied in this lab. Review them to deepen your understanding beyond just the configuration commands.

1. What is NAT Overload (PAT)?

Network Address Translation (NAT) Overload — also called **Port Address Translation (PAT)** — is the technology that allows an entire network of private IP addresses to share a single public IP address when accessing the internet. When a LAN host sends a packet to the internet, the router replaces the host's private source IP (e.g., 192.168.2.15) with its own WAN IP, but also assigns a unique source port number (e.g., port 54201) to track the session. When the internet server replies, the router uses that port number to determine which internal host the reply belongs to and forwards it correctly. This is why hundreds of devices in your home can all browse the internet simultaneously using just one public IP address assigned by your ISP.

2. What is a Standard ACL?

A **Standard Access Control List (ACL)** is a numbered or named list of rules that matches network traffic based solely on the **source IP address**. Standard ACLs use numbers 1–99 (and extended range 1300–1999). In this lab, ACL 1 permits all traffic sourced from the 192.168.2.0/24 subnet, which is then

referenced by the NAT overload command. A critical best-practice rule: standard ACLs should be **applied as close to the destination as possible**, because they can only match on source address — applying them near the source could block too much traffic unintentionally.

3. What is DHCP?

Dynamic Host Configuration Protocol (DHCP) automates the assignment of IP addresses, subnet masks, default gateways, and DNS server addresses to network hosts. Instead of manually configuring an IP on every device, a host sends a broadcast DHCP Discover message when it joins the network. The DHCP server (in this lab, the Cisco 1941 router) responds with an offer containing a lease — a temporary IP address from its defined pool. Key concepts: the **pool** defines the available address range, **exclusions** protect addresses reserved for infrastructure, and the **lease time** defines how long a host may use an IP before renewing the lease with the server.

4. WAN vs. LAN Interfaces — Inside vs. Outside Roles

In a routed NAT network, every interface must be assigned a NAT **role**. The **WAN interface (Gi0/0)** — connected to the modem and the internet — is designated `ip nat outside`. This tells IOS that this interface faces the "public" world. The **LAN interface (Gi0/1)** — connected to the switch and private hosts — is designated `ip nat inside`. This tells IOS that this interface faces the "private" network. These role designations are what allow IOS to know which direction to perform address translation: outbound traffic is translated from inside (private) to outside (public), and inbound replies are reverse-translated from outside back to inside.

5. Why ip nat inside / ip nat outside Matters

Without correctly applying `ip nat inside` and `ip nat outside` to the respective interfaces, NAT will simply **not function** — even if the NAT overload command and ACL are perfectly configured. Cisco IOS evaluates NAT only on packets that cross between an "inside" and an "outside" interface. If both interfaces are unconfigured, or if the roles are reversed, the router will forward packets without performing translation. This is one of the most common mistakes in NAT lab setups. Always verify with `show ip nat statistics` — it shows which interfaces are inside/outside and how many translations have been performed.

Lab prepared for personal practice — not for commercial distribution.

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 **Good luck with your studies, Surya!**

My Notes:
